Topic 2: Intelligent Agents Midterm Study Guide

CSE 412 / CSE 316 / CSE 411 | Artificial Intelligence | Daffodil International University

সহজ ইংরেজি + বাংলা | Beginner Level

Based on your uploaded files:
• Topic 2: Intelligent Agents (34 lecture slides)
• Midterms: Spring 2026, Fall 2025, Summer 2025, Spring 2025

Contents
Part 1 – Lecture in easy English + বাংলা (6 steps)
Part 2 – 3 past questions + 3 practice questions (Scenario → Question → Answer)
Part 3 – Colour-coded probability table + 5 exam tips

Lecture content comes from your slides. Past-question answers are written by me using standard AI knowledge, so please check them against your teacher's notes.

PART 1: Lecture 2 – Intelligent Agents (বুদ্ধিমান এজেন্ট)

Step 1. What is an Agent? (এজেন্ট কী?)

English: An agent **perceives** its environment with **sensors** and **acts** on that environment with its **actuators**. An agent gets percepts one at a time and maps this **percept sequence** to actions.

> **বাংলা:** এজেন্ট হলো এমন কিছু যা **সেন্সর** (যেমন চোখ, কান, ক্যামেরা) দিয়ে পরিবেশ দেখে এবং **অ্যাকচুয়েটর** (যেমন হাত, পা, চাকা, মোটর) দিয়ে পরিবেশে কাজ করে। সে একটি একটি করে percept (দেখা তথ্য) পায় এবং percept-এর তালিকাকে action-এ রূপ দেয়।

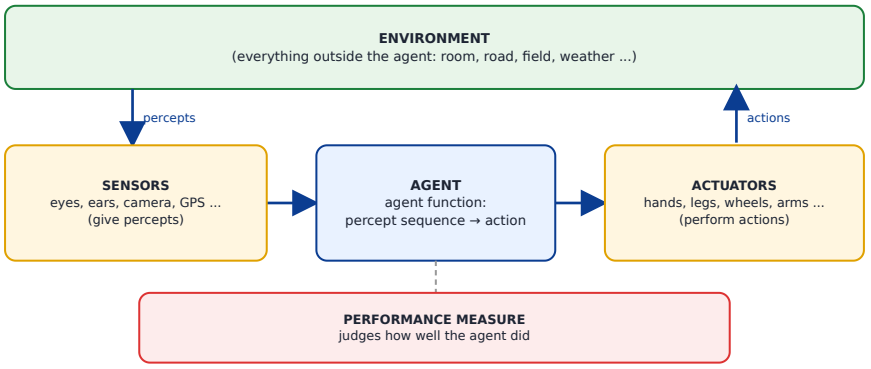


Figure 1: The agent–environment loop (এজেন্ট ও পরিবেশের চক্র)

Term	Meaning	বাংলা
Percept	What the agent senses at one moment	একটি মুহূর্তে এজেন্ট যা দেখে/অনুভব করে
Percept sequence	The complete history of everything the agent has ever perceived	এজেন্ট এ পর্যন্ত যা যা দেখেছে তার পুরো ইতিহাস

Step 2. Example: The Vacuum-Cleaner Agent (ভ্যাকুয়াম ক্লিনার এজেন্ট)

- The vacuum world has just **two squares: A and B**.
- The agent perceives **which square it is in** and **whether there is dirt**.
- It can choose to move **left**, move **right**, **suck** up dirt, or do nothing (**no-op**).
- Simple agent function: **if the current square is dirty, suck; otherwise move to the other square.**

Item	Value
Environment	Squares A and B
Percepts	[location, status], for example [A, Dirty]
Actions	Left, Right, Suck, No-op

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
    if status == Dirty then return Suck
    else if location == A then return Right
    else if location == B then return Left
```

> **বাংলা:** দুটি ঘর A ও B। এজেন্ট দেখে সে কোন ঘরে আছে এবং ঘরটি নোংরা কি না। নোংরা হলে **Suck** (ময়লা টানো); না হলে অন্য ঘরে যাও (A-তে থাকলে Right, B-তে থাকলে Left)।

Step 3. Rational Agent (যুক্তিসংগত এজেন্ট)

Definition (from slides): For each possible percept sequence, a rational agent should select an action that is expected to **maximize its performance measure**, given the evidence from the percept sequence and whatever **built-in prior knowledge** the agent has.

Term	Meaning	বাংলা
Performance measure	A way of calculating how well the agent has performed	এজেন্ট কতটা ভালো কাজ করল তা মাপার উপায়
Prior knowledge	Knowledge about the environment that the designer gave the agent before it started	কাজ শুরুর আগে ডিজাইনার যে জ্ঞান দিয়েছে
Agent function	Maps percept sequences to actions: `f : seq(P) → A`	percept-এর তালিকা থেকে action বের করার ফাংশন

বাংলা: যে এজেন্ট প্রতিটি পরিস্থিতিতে এমন action বেছে নেয় যা তার performance measure সবচেয়ে বেশি বাড়াবে বলে আশা করা যায়, সে rational agent। সে যা দেখেছে এবং আগে থেকে যা জানে, তার ভিত্তিতেই সিদ্ধান্ত নেয়।

Step 4. Task Environment and PEAS (টাস্ক এনভায়রনমেন্ট ও PEAS)

To design a rational agent, we must first specify its **task environment** using **PEAS**:

Letter	Meaning	বাংলা
Performance measure	Used to evaluate how well the agent solves the task	কাজ কতটা ভালো হলো তা মাপার মানদণ্ড
Environment	Surroundings beyond the control of the agent	এজেন্টের নিয়ন্ত্রণের বাইরের চারপাশ
Actuators	Used by the agent to perform actions	যা দিয়ে এজেন্ট কাজ করে
Sensors	Provide information about the current state of the environment	যা পরিবেশের তথ্য দেয়

Slide example: PEAS of an automated taxi (স্বয়ংক্রিয় ট্যাক্সি)

Performance	Safety, destination, profits, legality, comfort
Environment	Streets/motorways, other traffic, pedestrians, weather
Actuators	Steering, accelerator, brake, horn, speaker/display
Sensors	Video, sonar, speedometer, engine sensors, keyboard, GPS

Step 5. Environment Types (পরিবেশের ধরন)

An environment can be described by six pairs of properties. Memory trick: **O-D-E-S-D-S** (Observable, Deterministic, Episodic, Static, Discrete, Single-agent).

Property	Meaning (from slides)	বাংলা (সহজ কথায়)
Fully vs partially observable	Fully observable when the sensors can detect **all aspects relevant** to choosing an action.	কাজের জন্য দরকারি সব তথ্য সেন্সর দেখতে পেলে **fully**; কিছু লুকানো থাকলে **partially**।
Deterministic vs stochastic	Deterministic if the next state is **completely determined** by the current state and the action taken.	বর্তমান অবস্থা আর action দেখেই পরের অবস্থা নিশ্চিত বলা গেলে **deterministic**; দৈবের হাত (পাশা, ট্রাফিক) থাকলে **stochastic**।
Episodic vs sequential	Episodic: experience divides into atomic steps (perceive, then one action); the choice depends **only on the episode itself**.	প্রতিটি ধাপ আলাদা, আগের সিদ্ধান্তের প্রভাব নেই হলে **episodic**; আজকের সিদ্ধান্ত ভবিষ্যৎ বদলে দিলে **sequential**।
Static vs dynamic	Dynamic if the environment can **change while the agent is choosing** an action. Semi-dynamic if the environment stays the same but the agent's performance score changes with time.	এজেন্ট ভাবার সময় পরিবেশ বদলালে **dynamic**; পরিবেশ একই থাকে কিন্তু সময়ের সঙ্গে স্কোর বদলালে **semi-dynamic**।
Discrete vs continuous	Applies to the environment state, the way time is handled, and the percepts/actions.	সীমিত সংখ্যক অবস্থা/action হলে **discrete** (দাবার চাল); অসংখ্য মান (গতি, অবস্থান) হলে **continuous**।
Single vs multi-agent	Multi-agent if other agents are also **maximizing a performance measure that depends on this agent's actions**.	অন্য এজেন্টও নিজের লাভ বাড়াতে চায় এবং আমার action তাদের ফলাফলে প্রভাব ফেললে **multi-agent**।

Slide table: four example environments (চারটি উদাহরণ)

	Solitaire	Backgammon	Internet shopping	Taxi
Observable	FULL	FULL	PARTIAL	PARTIAL
Deterministic	YES	NO	YES	NO
Episodic	NO	NO	NO	NO
Static	YES	YES	SEMI	NO
Discrete	YES	YES	YES	NO
Single-agent	YES	NO	NO	NO

Remember: The **simplest** environment is fully observable, deterministic, episodic, static, discrete and single-agent. **Most real situations** are partially observable, stochastic, sequential, dynamic, continuous and multi-agent.

বাংলা: সবচেয়ে সহজ পরিবেশ: fully observable, deterministic, episodic, static, discrete, single-agent। বাস্তব জীবনের বেশিরভাগ পরিবেশ উল্টো: partially observable, stochastic, sequential, dynamic, continuous, multi-agent।

Step 6. Four Agent Types (চার ধরনের এজেন্ট)

Type	How it works (from slides)	Slide example	বাংলা
Simple reflex	Chooses an action using **only the current percept**. Uses **condition-action rules**: if dirty then suck.	Mail-sorting robot: city = Edin → put in Scotland bag. Also the vacuum agent.	শুধু বর্তমান percept দেখে "যদি-তাহলে" নিয়মে কাজ করে। স্মৃতি নেই।
Model-based reflex	For **partially observable** environments. Keeps an **internal state** built from percept history. Needs a model: how the world changes on its own, and how actions affect the world.	Robot vacuum cleaner with a map of the room and which areas are already cleaned.	যা দেখতে পায় না তা মনে রাখে। ভেতরে পৃথিবীর একটি মডেল (ম্যাপ) রাখে।
Goal-based	Has a **goal** to know which situations are desirable. Takes the **future** into account. Uses search and planning.	Robot maid: clean clothes, tidy room, table laid.	লক্ষ্য ঠিক করে ভবিষ্যৎ ভেবে কাজের ধারা বের করে।
Utility-based	A **utility function** maps states to a real number. Chooses between conflicting goals, or between goals using likelihood of success and importance.	Mars lander: shortest and obstacle-free path.	একাধিক উপায়ের মধ্যে কোনটি "বেশি ভালো" তা সংখ্যায় মেপে সেরাটি বাছে।

Quick rule to pick the type: No memory needed → simple reflex. Cannot see everything, needs memory/map → model-based. Must reach a target → goal-based. Many ways to reach it and some are better, or goals conflict → utility-based.
দ্রুত নিয়ম: স্মৃতি লাগে না → simple reflex; ম্যাপ/স্মৃতি লাগে → model-based; নির্দিষ্ট লক্ষ্য → goal-based; অনেক পথের মধ্যে সেরা বাছতে হলে → utility-based।

PART 2: Past Questions and Practice Questions (পূর্বের প্রশ্ন ও অনুশীলনী)

PAST = from your real exam papers. **PRACTICE** = extra practice. Each has **Scenario → Question → Answer (step by step)**.

PAST 1 AgriBot: PEAS and Environment Types (Fall 2025, Q1a and Q1b, 10 marks)

SCENARIO

AgriBot is an AI agent that boosts productivity, sustainability and farmer confidence. In Greenfield County, farmers struggle with unpredictable weather and pests. AgriBot scans the fields, checking soil moisture, crop health and pest activity. It gives eco-friendly solutions on damaged areas, and optimizes planting schedules and yields.

QUESTION

(a) Define the PEAS of AgriBot with a diagram. [5] (b) Explain the working model of the agentic solution, defining environment types with problem formulation criteria. [5]

ANSWER

Step 1 **(a) Draw the PEAS diagram.**

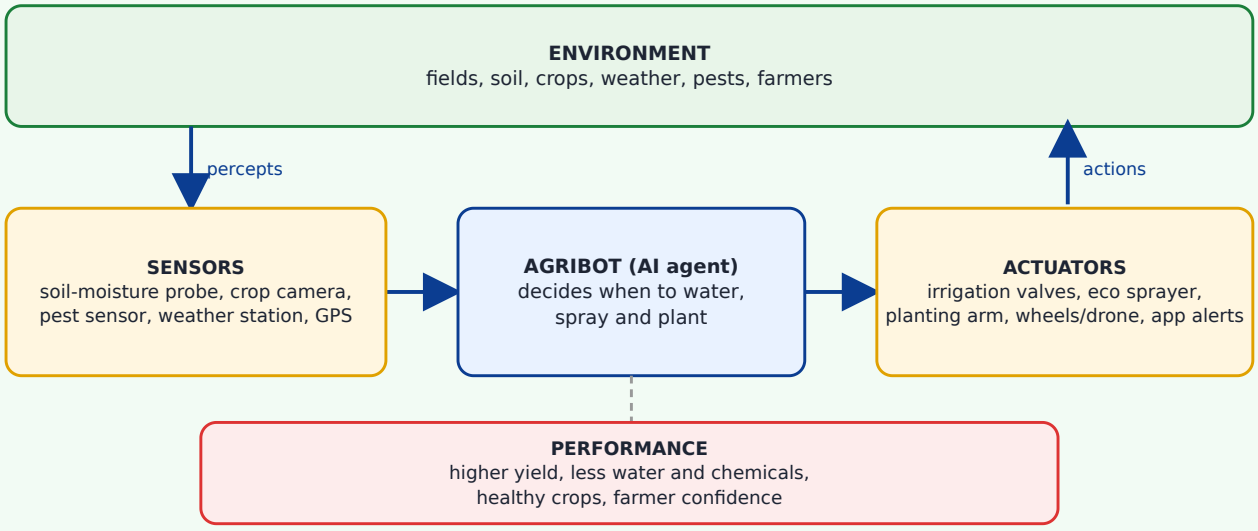


Figure 2: PEAS diagram of AgriBot

Step 2 **Write the PEAS table.**

Performance	Higher crop yield, correct pest/disease detection, less water and chemicals used, healthy crops, farmer confidence
Environment	Fields, soil, crops, weather (rain, heat), pests, farmers
Actuators	Irrigation valves, eco-friendly sprayer, planting arm/seeder, wheels or drone motors, alerts to the farmer's app
Sensors	Soil-moisture probe, crop-health camera, pest sensor, weather station (temperature, humidity, rain), GPS

Step 3 **(b) Working model (sense → think → act loop).** (1) **Sense:** read soil, crop, pest and weather data. (2) **Update model:** keep a map of the field and the history of moisture and pests. (3) **Decide:** pick the best action to reach the goal (healthy crop, high yield) with the least resources. (4) **Act:** water, spray only damaged areas, adjust planting schedule. (5) **Feedback:** sense again and improve.

Step 4 **Environment types.**

Property	Answer	Reason
Observable	Partially	Cannot see every plant, root or hidden pest at once
Deterministic	Stochastic	Weather and pests are unpredictable
Episodic	Sequential	Today's watering and planting affect future yield
Static	Dynamic	Weather, moisture and pests change while the agent decides
Discrete	Continuous	Moisture, temperature and position take any value
Single-agent	Single-agent	Only AgriBot is acting to optimize the farm (farmers only supervise)

Step 5 **Problem formulation criteria.**

Criterion	For AgriBot
Initial state	Current soil moisture, crop health and pest levels of the field
Actions	Water, spray, plant, wait, alert
Transition model	How each action changes moisture, pests and crop health
Goal test	Crop healthy and yield target reached
Path cost	Water, chemicals, energy and time used

Agent type: goal-based, or utility-based because it balances yield against resource use.

বাংলা: AgriBot মাটির আর্দ্রতা, ফসলের স্বাস্থ্য ও পোকা সেন্সর দিয়ে দেখে, তথ্য মনে রাখে, সবচেয়ে কম পানি/রাসায়নিকে ভালো ফলন পাওয়ার সিদ্ধান্ত নেয় এবং সেচ/স্প্রে করে। পরিবেশ: partially observable, stochastic, sequential, dynamic, continuous, single-agent। এটি goal-based বা utility-based agent।

PAST 2 Swarm Robots: PEAS, Environment and State of the Art (Spring 2025, Q1, 8 marks)

SCENARIO

An environmental monitoring system uses bio-sensor enabled **swarm robots**. This autonomous AI system works as an intelligent agent on toxic gases and pathogens in the environment, concerned with model computation time.

QUESTION

(a) Discuss PEAS using this case study. [2] (b) Develop an AI system considering environment types with an agent and demonstrate its functionalities. [3] (c) Explain the state of the art of AI using a problem-solving agent. [3]

ANSWER

Step 1 **(a) PEAS.**

Performance	Detection accuracy, fast response, area covered, few false alarms, battery life, low computation time
Environment	Air and area to be monitored, toxic gases, pathogens, wind, temperature, obstacles, other robots
Actuators	Wheels or propellers, radio transmitter (to talk to the swarm), alarm/alert unit, sampling pump
Sensors	Gas sensors, bio-sensors, temperature/humidity sensors, GPS, proximity sensors

Step 2 **(b) Environment types. Partially observable** (each robot senses only its nearby air); **stochastic** (wind spreads gas unpredictably); **sequential** (earlier moves decide coverage); **dynamic** (gas concentration keeps changing); **continuous** (concentration and position are real numbers); **multi-agent** (many cooperating robots).

Step 3 **(b) The AI system and its functions.** Each robot is a **model-based, utility-based agent**: it keeps a map of readings and chooses moves that give the most information.

1. **Sense:** read gas and bio-sensor values and location.
2. **Share:** send readings to nearby robots (swarm communication).
3. **Update model:** build a shared hazard map.
4. **Decide:** move toward higher concentration to find the source, while spreading out to cover the area.
5. **Act:** move, sample, raise an alarm if a threshold is crossed.
6. **Save time:** use simple local rules on each robot, so computation stays fast.

Step 4 **(c) State of the art of AI.** From the lecture, AI is used today in: autonomous planning and scheduling, game playing, autonomous control, medical diagnosis, logistics planning, robotics, and language understanding and problem solving.

Step 5 **(c) Problem-solving agent steps**, shown on the swarm example:

Step	Meaning	Swarm example
1. Goal formulation	Decide what to achieve	Find the source of the toxic gas
2. Problem formulation	Define states, actions, goal test, cost	State = robot positions and readings; actions = move/sample; cost = time and battery
3. Search	Find a sequence of actions that reaches the goal	Search for the best movement plan
4. Execute	Carry out the plan, watch the percepts	Move, sense again, re-plan if wind changes

বাংলা: প্রতিটি রোবট গ্যাস ও বায়ো-সেন্সর দিয়ে তথ্য নেয়, অন্য রোবটের সঙ্গে ভাগ করে, ঝুঁকির ম্যাপ বানায়, গ্যাসের উৎসের দিকে যায় এবং সীমা পার হলে অ্যালার্ম দেয়। পরিবেশ: partially observable, stochastic, sequential, dynamic, continuous, multi-agent। Problem-solving agent-এর ৪ ধাপ: goal formulation, problem formulation, search, execute।

PAST 3 Maze: Classify the Task Environment (Spring 2026, Q2a, 5 marks)

SCENARIO

An intelligent system must solve a maze. The maze has an entrance on the left and an exit on the right, with many walls and dead ends (Figure 2(a) of the exam, "Example puzzle game board").

QUESTION

Classify the task environment and describe the problem-solving steps for the intelligent system solving the maze.

ANSWER

Step 1 **Classify the task environment.**

Property	Answer	Reason
Observable	Fully	The whole maze map is visible to the system (if it could only sense nearby walls, it would be partially observable)
Deterministic	Deterministic	A move always gives the expected next cell
Episodic	Sequential	Each move changes the position and affects later moves
Static	Static	Walls do not change while the system thinks
Discrete	Discrete	Finite number of cells and four moves
Single-agent	Single-agent	No other agent affects the result

Step 2 **Problem-solving steps.**

Step	For the maze
1. Goal formulation	Reach the exit on the right side
2. Problem formulation	**States:** the cell the agent is in. **Initial state:** the entrance cell. **Actions:** up, down, left, right (only if there is no wall). **Transition:** moves to the neighbouring cell. **Goal test:** is the agent at the exit? **Path cost:** 1 per move.
3. Search	Use a search algorithm such as BFS (shortest path) or A* (with a distance heuristic) to find a path of cells from the entrance to the exit
4. Execute	Follow the path move by move to the exit

বাংলা: গোলকধাঁধার ম্যাপ পুরোটা দেখা যায়, তাই fully observable। প্রতিটি চালের ফল নিশ্চিত (deterministic), দেয়াল বদলায় না (static), ঘরের সংখ্যা সীমিত (discrete), আর একজনই এজেন্ট (single-agent)। সমাধানের ৪ ধাপ: লক্ষ্য ঠিক করা (বের হওয়ার পথ), সমস্যা সাজানো (state, action, goal test, cost), search চালানো (BFS/A*), এবং পথ ধরে চলা।

PRACTICE 1 Vacuum-World Agent Function and Rationality (from the lecture example)

SCENARIO

A vacuum cleaner works in a world with two squares, A and B. Each percept is [location, status], where status is Clean or Dirty. The actions are Left, Right, Suck and No-op.

QUESTION

(a) Write the agent function as a table. (b) Write the agent program. (c) Is this agent rational? Justify.

ANSWER

Step 1 **(a) Agent function table** (maps a percept to an action):

Percept [location, status]	Action
[A, Dirty]	Suck
[A, Clean]	Right
[B, Dirty]	Suck
[B, Clean]	Left

Step 2 **(b) Agent program:**

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
    if status == Dirty then return Suck
    else if location == A then return Right
    else if location == B then return Left
```

Step 3 **(c) Is it rational?** It depends on the performance measure. Suppose the measure is "1 point for each clean square at each time step". Then the agent is rational, because it always cleans dirt and visits both squares. If every move costs energy (a penalty), it is **not** rational: after both squares are clean it keeps moving back and forth. A rational agent would then choose **No-op**. This agent is a **simple reflex agent**, because it uses only the current percept.

বাংলা: Rational কি না তা নির্ভর করে performance measure-এর ওপর। "প্রতি সময়ে পরিষ্কার ঘরের জন্য ১ পয়েন্ট" হলে এজেন্ট rational। কিন্তু প্রতিটি চালে শক্তি খরচ হলে দুই ঘর পরিষ্কার হওয়ার পরও সে ঘুরতে থাকে, তাই rational নয়; তখন No-op করা উচিত।

PRACTICE 2 Classify Four Environments

SCENARIO

You are asked to design agents for four tasks: a crossword-puzzle solver, a chess player with a game clock, a poker player, and a medical-diagnosis assistant.

QUESTION

Classify each task environment on all six properties and name the simplest and the most complex one. Justify briefly.

ANSWER

Step 1 **Fill the table.**

	Crossword	Chess with clock	Poker	Medical diagnosis
Observable	Full	Full	Partial (hidden cards)	Partial
Deterministic	Yes	Yes (apart from the opponent's choice)	No (cards are random)	No
Episodic	No (sequential)	No (sequential)	No (sequential)	No (sequential)
Static	Static	Semi-dynamic (clock runs)	Static	Dynamic (patient changes)
Discrete	Yes	Yes	Yes	No (continuous values)
Single-agent	Yes	No (opponent)	No (opponents)	Yes

Step 2 **Conclusion.** The **crossword** is the simplest (closest to fully observable, deterministic, static, discrete, single-agent). **Medical diagnosis** is the most complex and most like real life (partially observable, stochastic, sequential, dynamic, continuous).

বাংলা: ক্রসওয়ার্ড সবচেয়ে সহজ পরিবেশ। মেডিকেল ডায়াগনোসিস সবচেয়ে জটিল, কারণ রোগীর সব তথ্য দেখা যায় না, ফল অনিশ্চিত, অবস্থা বদলায় ও মানগুলো continuous। পোকারে প্রতিপক্ষের কার্ড লুকানো, তাই partially observable।

PRACTICE 3 Choose the Right Agent Type

SCENARIO

Five systems are given: (1) a mail-sorting robot on a conveyor belt, (2) a robot vacuum that remembers a map of the room, (3) a robot maid that must clean clothes, tidy the room and lay the table, (4) a Mars lander that must find a short, obstacle-free path, (5) a smart thermostat that turns the heater on when the temperature is below 20 degrees.

QUESTION

Name the best agent type for each system and justify your choice.

ANSWER

Step 1 **Match and justify.**

#	System	Agent type	Justification
1	Mail-sorting robot	**Simple reflex**	Decides from the current letter only: if city = Edin then put in the Scotland bag. No memory needed.

2	Robot vacuum with room map	**Model-based reflex**	It cannot see the whole room, so it keeps an internal state (map, cleaned areas) built from percept history.
3	Robot maid	**Goal-based**	It has goals (clean clothes, tidy room, table laid) and must plan a long sequence of actions, thinking about the future.
4	Mars lander	**Utility-based**	Many paths reach the target, but some are better. A utility function scores shortest and obstacle-free paths, and the lander picks the highest.
5	Smart thermostat	**Simple reflex**	Single condition-action rule: if temperature < 20 then heater on.

Step 2 **Memory line:** Reflex = "react now"; Model-based = "remember"; Goal-based = "plan to reach"; Utility-based = "choose the best".

বাংলা: চিঠি বাছাই রোবট ও থার্মোস্ট্যাট: শুধু বর্তমান অবস্থা দেখে নিয়মে কাজ করে (simple reflex)। রোবট ভ্যাকুয়াম: ঘরের ম্যাপ মনে রাখে (model-based)। রোবট মেইড: লক্ষ্য ঠিক করে পরিকল্পনা করে (goal-based)। মঙ্গল ল্যান্ডার: সেরা পথ সংখ্যায় মেপে বাছে (utility-based)।

PART 3: Probability Table and Exam Tips (সম্ভাবনা ছক ও পরীক্ষার টিপস)

These probabilities are MY ESTIMATE from the 4 past papers you uploaded. They are NOT a guarantee. (এগুলো আমার অনুমান, নিশ্চয়তা নয়।)

Step 1. Colour-coded probability table (Topic 2)

Chance	Topic	Seen in	Why
VERY HIGH	PEAS of a given scenario (with diagram)	Fall 25, Spring 25	Appears as the opening question; every scenario needs PEAS
VERY HIGH	Environment types (classify with reasons)	Fall 25, Spring 25, Spring 26	3 of 4 papers
HIGH	Define intelligent agent; type of agent with justification	Spring 26	Direct 5-mark question; the four types are easy to test
HIGH	Four agent types with examples	Spring 26 (type asked)	Easy theory plus scenario matching
HIGH	Problem formulation and problem-solving agent steps	Fall 25, Summer 25, Spring 25, Spring 26	Links Topic 2 to search questions
MEDIUM	Rational agent, performance measure, percept sequence	Not asked directly	Core theory; can appear as a short definition
MEDIUM	Vacuum-world agent function and program	Not seen	Slide example; possible 2-3 mark part
MEDIUM	Agent–environment diagram (sensors, actuators)	Fall 25 (diagram)	Often part of a PEAS or agent answer

লাল = খুব বেশি সম্ভাবনা, কমলা = বেশি, হলুদ = মাঝারি।

Step 2. Five exam tips (পরীক্ষার ৫টি টিপস)

#	Tip (English)	টিপস (বাংলা)
1	**Start every scenario with a PEAS table.** Write 3 to 4 items in each of P, E, A, S, using words from the scenario.	**প্রতিটি দৃশ্যপটে আগে PEAS ছক লিখুন।** P, E, A, S প্রতিটিতে ৩-৪টি পয়েন্ট, প্রশ্নের শব্দ ব্যবহার করে।
2	**Give all 6 environment types with a one-line reason.** Never write only the name; the reason earns the marks.	**৬টি পরিবেশ-ধরনই কারণসহ লিখুন।** শুধু নাম লিখলে নম্বর কমে; কারণই নম্বর দেয়।
3	**Name the agent type, then justify with 2 reasons.** Say why a simpler type is not enough (for example, "needs a map, so not simple reflex").	**এজেন্টের ধরন বলে ২টি কারণ দিন।** কেন সহজ ধরন যথেষ্ট নয় তাও লিখুন।
4	**Draw the diagram.** A sensors → agent → actuators → environment loop is quick to draw and the paper may ask for it.	**ডায়াগ্রাম আঁকুন।** সেন্সর → এজেন্ট → অ্যাকচুয়েটর → পরিবেশ চক্র দ্রুত আঁকা যায়, প্রশ্নে চাইতেও পারে।
5	**Manage time and label parts.** 25 marks in 1.5 hours is about 3.5 minutes per mark. Answer a), b), c) in order.	**সময় ভাগ করুন ও অংশ ক্রমে লিখুন।** প্রতি নম্বরে প্রায় ৩.৫ মিনিট। a), b), c) ক্রমানুসারে লিখুন।